Last updated: August 6, 2026

# KIHYUN YU

**Address**
Industrial and Systems Engineering,
KAIST
291 Daehak-ro, Yuseong-gu, Daejeon, Republic of Korea

**Email** khyu99@kaist.ac.kr
**Website** https://kihyun-yu.github.io

## RESEARCH INTERESTS

My research focuses on **safe reinforcement learning (RL) theory**. In particular, I design algorithms for **constrained MDPs** with provable guarantees under various settings; adversarial, non-stationary, function approximation, and infinite-horizon formulations. I am also broadly interested in **sequential decision-making**, including **bandit algorithms**, **online convex optimization**, and their real-world applications.

## EDUCATION

- KOREA ADVANCED INSTITUTE OF SCIENCE AND TECHNOLOGY (KAIST) — South Korea
  - Integrated M.S.–Ph.D. Program in Industrial and Systems Engineering — 03/2024 – present
  - Advisor: Prof. Dabeen Lee
  - GPA: 4.2 / 4.3
- GWANGJU INSTITUTE OF SCIENCE AND TECHNOLOGY (GIST) — South Korea
  - B.S. in Electrical Engineering and Computer Science — 03/2018 – 02/2024
  - Minor in Mathematics
  - GPA: 4.08 / 4.5 (*cum laude*)
  - Two-year leave for mandatory military service (2020 – 2021)
- SEJONG SCIENCE HIGH SCHOOL — South Korea
  03/2015 – 02/2018

## PUBLICATIONS

### PREPRINTS

[1] **Kihyun Yu**, Junehee Lee, Dabeen Lee. Constrained Online Convex Optimization without Slater's Condition. Available at: https://arxiv.org/abs/2606.31480

[2] **Kihyun Yu**, Beomhan Baek, Dabeen Lee. Learning Weakly Communicating Average-Reward CMDPs: Strong Duality and Improved Regret. Available at: https://arxiv.org/abs/2605.11586

### PUBLICATIONS

[3] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Near-Optimal Primal-Dual Algorithm for Learning Linear Mixture CMDPs with Adversarial Rewards. *IEEE Conference on Decision and Control (CDC), 2026.* Available at: https://arxiv.org/abs/2603.27884

[4] **Kihyun Yu**, Seoungbin Bae, Dabeen Lee. Primal-Dual Policy Optimization for Linear CMDPs with Adversarial Losses. *International Conference on Learning Representations (ICLR)*, 2026. Available at: https://arxiv.org/abs/2605.11535

[5] Jiahui Zhu, **Kihyun Yu**, Dabeen Lee, Xin Liu, Honghao Wei. An Optimistic Algorithm for online CMDPs with Anytime Adversarial Constraints. *International Conference on Machine Learning (ICML), 2025.* Available at: https://arxiv.org/abs/2505.21841

[6] **Kihyun Yu**, Duksang Lee, William Overman, and Dabeen Lee. Improved Regret Bound for Safe Reinforcement Learning via Tighter Cost Pessimism and Reward Optimism. *Reinforcement Learning Conference (RLC), 2025.* Available at: https://rlj.cs.umass.edu/2025/papers/Paper41.html

## Academic Services

- Conference Reviewer: ICML 2026 (silver reviewer), NeurIPS 2026, IEEE CDC 2026, AAAI 2027
- Journal Reviewer: TMLR, IEEE Transactions on Networking

## Teaching Experience

- Teaching Assistant, Operations Research: Optimization (KAIST) — 03/2024 – 06/2024
- Teaching Assistant, Differential Equations (GIST) — 09/2023 – 12/2023
- Teaching Assistant, Single Variable Calculus (GIST) — 03/2023 – 06/2023

## Honors

- Scholarship for Academic Excellence
  - Awarded for 6 semesters (Spring 2018, Fall 2018, Spring 2019, Fall 2019, Spring 2022, Spring 2023)

## Experience

- Information Processing, Controlling, and Network Lab (GIST) — South Korea
  - Undergraduate Research Intern (Advisor: Prof. Heung-No Lee) — 01/2023 – 12/2023
  - Conducted research on lattice-based cryptography
- Artificial Intelligence Lab (GIST) — South Korea
  - Undergraduate Research Intern (Advisor: Prof. Kyoobin Lee) — 07/2022 – 12/2022
  - Conducted research on robust image classification under noisy labels
- University of Cambridge — United Kingdom
  - Summer Session Programme (Machine Learning) — 07/2019 – 08/2019

## Languages and Technical Skills

- Korean (native), English (conversational)
- C/C++, Python, PyTorch, LaTeX

## Other Activities

- Volunteer Math Mentor — 03/2018 – 12/2018
  - Provided small-group math mentoring for middle and high school students